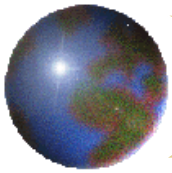
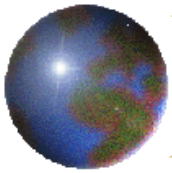


Chapter 14
Wiener Processes and Itô's Lemma



Stochastic Processes

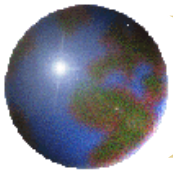
- Describes the way in which the value of a variable such as a stock price, exchange rate or interest rate changes over time.
- Incorporates uncertainties
- Examples of stochastic process include the Wiener process, Brownian motion process, Poisson process etc.



Example

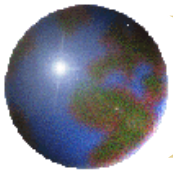
- Each day a stock price
  - increases by $1 with probability 30%
  - stays the same with probability 50%
  - reduces by $1 with probability 20%

- Stochastic processes can be classified as,
  - Discrete-time *(where the value of the variable can change only at certain fixed points in time,* or continuous-time *(where changes can take place at any time)*
  - *Continuous -variable (the underlying variable can take any value within a certain range* or *discrete-variable (only certain discrete values are possible).*



Modeling Stock Prices

- We can use any of the four types of stochastic processes to model stock prices
- However, in practice, we do not observe stock prices following continuous-variable, continuous-time processes (so complicated).
- Stock prices are restricted to discrete values (e.g., multiples of a cent) and changes can be observed only when the exchange is open for trading.
- Nevertheless, the continuous-time, the continuous - variable process proves to be a useful model for many purposes.



Markov Property

- A stochastic process has the *Markov property* if the conditional probability distribution of future states of the process (conditional on both past and present states) depends only upon the present state, not on the sequence of events that preceded it.
- A process with this property is called a *Markov process*.
- In a Markov process future movements in a variable depend only on where we are, not the history of how we got to where we are. Past events are irrelevant.
- Stock prices are usually assumed to follow a Markov process.

- For example, suppose that the current price of a stock is $100.
- If the stock price follows a Markov process, our predictions for the future movement of the stock price should be unaffected by the price one week ago, one month ago, or one year ago.
- The only relevant piece of information is that the price is now $100
- Predictions for the future are uncertain and must be expressed in terms of probability distributions.
- The Markov property implies that the probability distribution of the price at any particular future time is not dependent on the particular path followed by the price in the past.

Weak-Form Market Efficiency (Theory)

- The Markov property of stock prices is consistent with the weak form of market efficiency theory.
- This theory states that the present price of a stock impounds all the information contained in a record of past prices. So stock trades at the fairest value
- As such it is impossible to produce consistently superior returns with a trading rule based on the past history of stock prices.
- In other words technical analysis can not be effectively utilized to aid investors in making trading decisions.

Example

- Consider a variable that follows a Markov stochastic process.
- Suppose that its current value is 10 and that the change in its value during a year is $\Phi(0, 1)$, where $\Phi(\mu, v)$ denotes a probability distribution that is normally distributed with mean μ and variance v.
- The process is stationary (i.e. the parameters of the process do not change as we move through time)
- At the end of 1 year the variable will have a normal probability distribution with mean 0 and variance 1.0

Variances & Standard Deviations

- In Markov processes changes in the value of a variable in successive periods of time are independent
- This means that variances of the changes are additive and;
- Standard deviations of the changes are not additive

Variances & Standard Deviations
(continued)

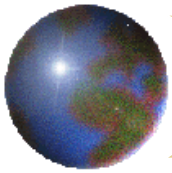
- The variance of the change in the variable in our example is 1.0 per year, in 2 years is 2.0 and in 3 years is 3.0.

A Wiener Process

- It is a particular type of Markov stochastic process with a mean change of zero and a variance rate of 1.0 per year.
- Define $\phi(\mu, v)$ as a normal distribution with mean μ and variance v
- A variable z follows a Wiener process if
  - *The change in z in a small interval of time Δt is Δz*

    $\Delta z = \varepsilon\sqrt{\Delta t}$ where ε is $\phi(0,1)$

  - *The values of Δz for any 2 different (non-overlapping) short intervals of time Δt are independent*



Properties of a Wiener Process

Follows from the first property that-

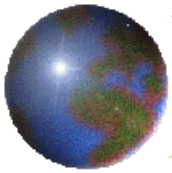
- Δz itself has a normal distribution with
  - Mean of $\Delta z = 0$
  - Variance of $\Delta z = \Delta t$
  - Standard deviation of $\Delta z = \sqrt{\Delta t}$

Properties of a Wiener Process….

- The second property implies that *z* follows a Markov process.
- For a relatively long period of time, *T*, it can be denoted by *z(T) – z (0)* that is normally distributed with
  - Mean of $[z(T) - z(0)]$ is 0
  - Variance of $[z(T) - z(0)]$ is T
  - Standard deviation of $[z(T) - z(0)]$ is $\sqrt{T}$

Generalized Wiener Processes

- A Wiener (stochastic) process has a drift rate (i.e. average/mean change per unit of time) of *0* and a variance rate (per unit of time) of *1.0*
- The basic Wiener process, *dz* that has been developed so far has a drift rate of *0* and a variance rate of 1.0.
- In a generalized Wiener process the drift rate and the variance rate can be set equal to any chosen constants



- The drift rate of *0* means that the expected value of *z* at any future time is equal to its current value.
- The variance rate of 1.0 means that the variance of the change in *z* in a time interval of length *T* equals *T*.

Generalized Wiener Processes
(continued)

- A *generalized Wiener process* for a variable x can be defined in terms of dz as

$$dx = a\,dt + b\,dz$$

- Where,
  - The mean change in x per unit time is a
  - Variance of change in x per unit time is b^2

- In a small time interval Δt, the change in the value of x is given by

$$\Delta x = a\ \Delta t + b\ \varepsilon\sqrt{\Delta t}$$

- Thus Δx has a normal distribution with
  - mean of Δx = *a* Δt
  - variance of Δx = b^2 Δt
  - standard deviation of Δx = b $\sqrt{}$Δt
- Similarly the change in the value of x in any time interval *T* is normally distributed with
  - mean of Δx = a*T*
  - variance of Δx = b^2 *T*
  - standard deviation of Δx = b $\sqrt{}$T

Figure 14.1 How a Wiener process is obtained when $\Delta t \to 0$ in equation (14.1).

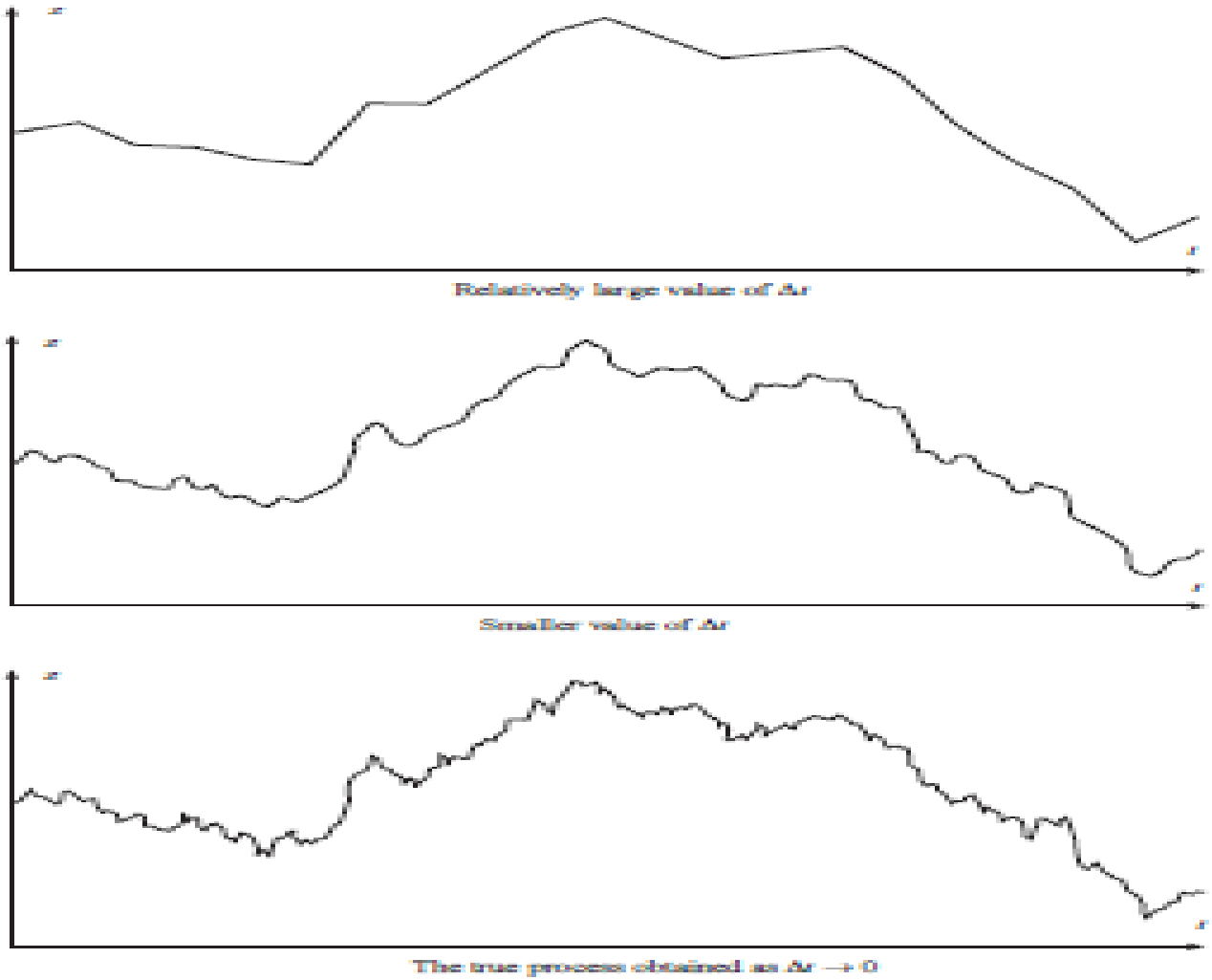
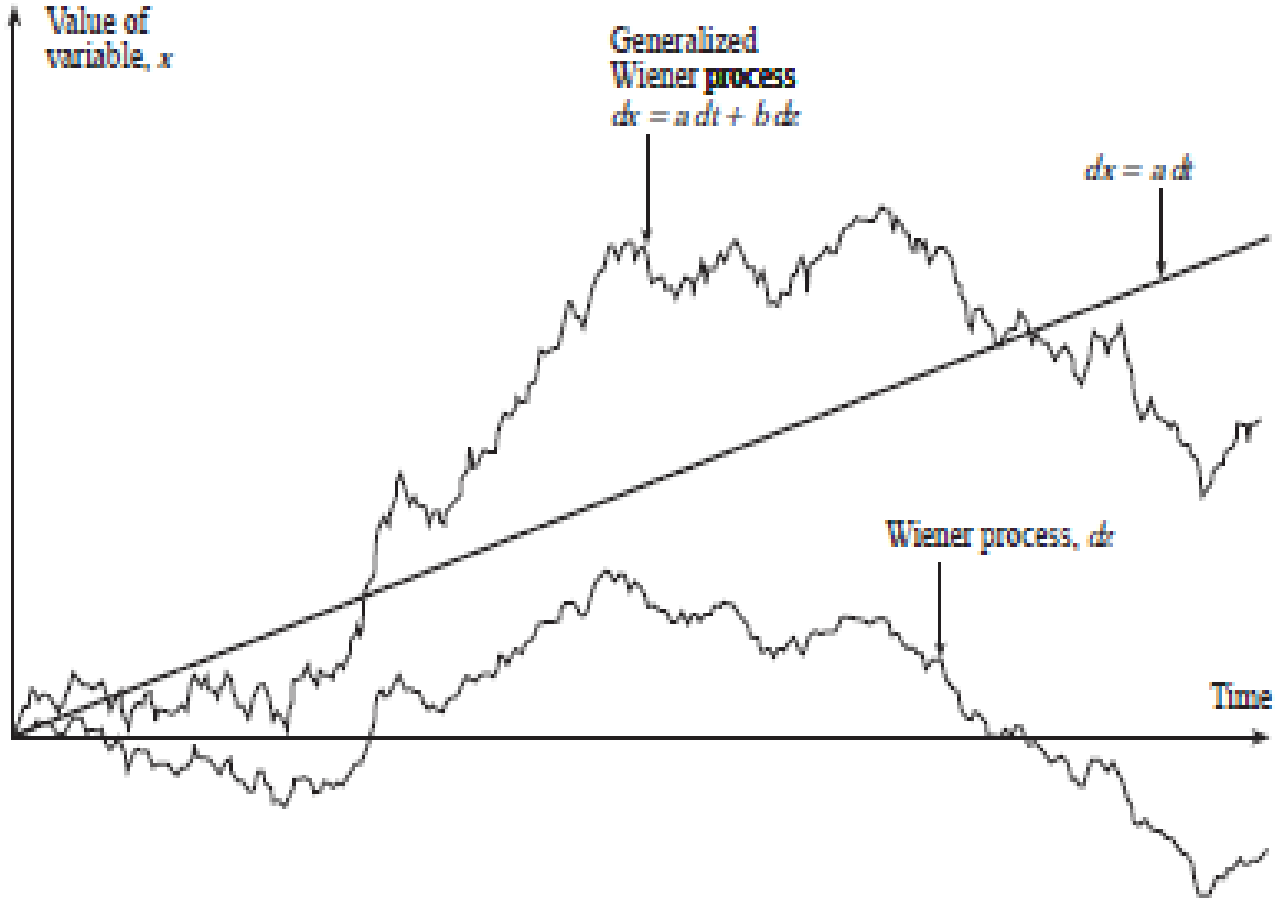


Figure 14.2 Generalized Wiener process with $a = 0.3$ and $b = 1.5$.



Example

- A stock price starts at 40 and has a probability distribution of $\phi(40,100)$ at the end of the year
- If we assume the stochastic process is Markov with no drift, then the process is

$$dS = 10dz$$

- If the stock price were expected to grow by \$8 on average during the year, so that the year-end distribution is $\phi(48,100)$, the process would be

$$dS = 8dt + 10dz$$

Itô Process

- A further type of stochastic process.
- This is a *generalized Wiener process* in which the drift rate and the variance rate are functions of the value of the underlying variable x and time t written as

$$dx = a(x, t)\,dt + b(x, t)\,dz$$

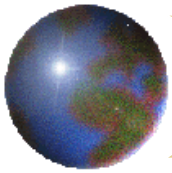
- Both the expected drift rate and variance rate of an Itoˆ process are liable to change over time.
- The discrete time equivalent

$$\Delta x = a(x,t)\Delta t + b(x,t)\varepsilon\sqrt{\Delta t}$$

is only true in the limit as Δt tends to zero.

The Process for a Stock Price

- It is attractive to suggest that a stock price (non-dividend paying stock) follows a generalized Wiener process; that is, it has a constant expected drift rate and a constant variance rate.
- However, this model fails to capture a key aspect of stock prices. That is the expected percentage return required by investors from a stock is independent of the stock's price
- If investors require a 14% per annum expected return when the stock price is $10, then, *ceteris paribus*, they will also require a 14% per annum expected return when it is $50.



- Thus clearly, the assumption of constant expected drift rate is inappropriate and needs to be replaced by the assumption that the expected return is constant
- If S is the stock price at time t, then the expected drift rate in S should be assumed to be μS for some constant parameter μ.
- This means that in a short interval of time,Δt, the expected increase in S is $\mu S \Delta t$.
- The parameter μ is the expected rate of return on the stock.

- If the coefficient of dz (second part of the equation) is zero, so that there is no uncertainty, then this model implies that
  - $\Delta S = \mu S \Delta t$, in the limit as $\Delta t \rightarrow 0$,
  - $dS = \mu S dt$
  - $dS/S = \mu dt$
- Integrating between 0 and T, we get
  - $S_T = S_0 e^{\mu T}$
- This equation shows that when there is no uncertainty, the stock price grows at a continuously compounded rate of μ per unit of time.

- In practice, of course, there is uncertainty.
- Reasonable to assume that the variability of the return in a short period of time is the same regardless of the stock price that is when the stock price is $50 as when it is $10.
- This suggests that the standard deviaton of the change in a short period of time should be proportional to the stock price and leads to the model

$$dS = \mu S\,dt + \sigma S\,dz$$

- Or $ds/s = \mu\,dt + \sigma\,dz$
- The above equation is the most widely used model of stock price behavior and represents the stock price process in the real world.

An Ito Process for Stock Prices

$$dS = \mu S\,dt + \sigma S\,dz$$

where μ is the expected return σ is the volatility.

The process is known as *geometric Brownian motion.*

The discrete time version of the model or its equivalent is

$$\Delta S = \mu S \Delta t + \sigma S \varepsilon \sqrt{\Delta t}$$

- Or
  $\Delta S/S = \mu\Delta t + \sigma\varepsilon\sqrt{\Delta t}$
- It shows that $\Delta S/S$ is normally distributed with mean $\mu\Delta t$ and standard deviation of $\sigma\sqrt{\Delta t}$. In other words,

$$\Delta S/S \sim \Phi(\mu\Delta t, \sigma^2\Delta t),$$

- Example:
- Consider a stock that pays no dividends, has a volatility of 30% per annum, and provides an expected return of 15% per annum with continuous compounding. The process for the stock price is

$$\frac{dS}{S} = 0.15\,dt + 0.30\,dz$$

- Small time interval: $\dfrac{\Delta S}{S} = 0.15\Delta t + 0.30\epsilon\sqrt{\Delta t}$
- For a time interval of 1 week (0.0192 years), it is,

$$\frac{\Delta S}{S} = 0.15 \times 0.0192 + 0.30 \times \sqrt{0.0192}\,\epsilon$$

$$\Delta S = 0.00288S + 0.0416S\epsilon$$

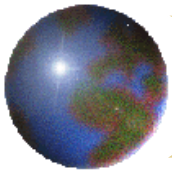
Monte Carlo Simulation

- A Monte Carlo simulation of a stochastic process is a procedure for sampling random outcomes for the process.
- We can sample random paths for the stock price by sampling values for ε
- Suppose as before, $\mu = 0.15$, $\sigma = 0.30$, and Δt = 1 week (=1/52 or 0.0192 years), then

$$\Delta S = 0.15 \times 0.0192 S + 0.30 \times \sqrt{0.0192}\varepsilon$$

or

$$\Delta S = 0.00288 S + 0.0416 S \varepsilon$$



Monte Carlo Simulation – One Sampling Path

Week	Stock Price at Start of Period	Random Sample for ε	Change in Stock Price, ΔS
0	100.00	0.52	2.45
1	102.45	1.44	6.43
2	108.88	−0.86	−3.58
3	105.30	1.46	6.70
4	112.00	−0.69	−2.89
5	109.11	-0.74	-3.04
.	.	.	.
.	.	.	.
10	111.54	2.56	12.20

- From the equation below:

$$\Delta S = 0.00288S + 0.0416S\varepsilon$$

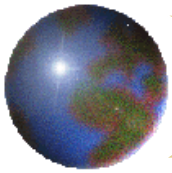
- The change during the first time period is

$$\Delta S = 0.00288 \times 100 + 0.0416 \times 100 \times 0.52 = 2.45$$

- The change during the second time period is

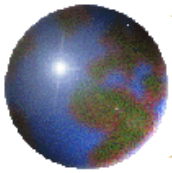
$$\Delta S = 0.00288 \times 102.45 + 0.0416 \times 102.45 \times 1.44 = 6.43$$

- And so on.



The Parameters

- Stock price process has two parameters, μ and σ.
- Parameter μ is the expected return (annualized) an investor in a short period of time earns.
- Higher expected returns depends on the level of risk involves.
- Returns also depend on the level of interest rates in the economy.
- The higher the level of interest rates, the higher the expected return required by an investor on any given stock.



- The value of a derivative is dependent on a the price of underlying asset. It is independent of μ.
- The parameter σ is, by contrast, critically important to the determination of the value of many derivatives.
- Standard deviation is a measure of the amount of variation or dispersion of a set of value (of a random variable for example)
- Typical values of σ for a stock are in the range 0.15 to 0.60 (i.e., 15% to 60%).

Itô's Lemma

- Generally, the price of any derivative (e.g. stock option) is a function of the stochastic variables underlying the derivative and time.

Itô's Lemma

- Generally, the price of any derivative (e.g. stock option) is a function of the stochastic variables underlying the derivative and time.
- If we know the stochastic process followed by a variable x, Itô's lemma tells us the stochastic process followed by a function $G\ (x,\ t\ )$, when $dx=a(x,t)\ dt+b(x,t)\ dz$ then follows the process

$$dG=\left(\frac{\partial G}{\partial x}a+\frac{\partial G}{\partial t}+\tfrac{1}{2}\frac{\partial^2 G}{\partial x^2}b^2\right)dt+\frac{\partial G}{\partial x}b\ dz$$

- Since a derivative is a function of the price of the underlying asset and time, Itô's lemma plays an important part in the analysis of derivatives.

Application of Itô's Lemma to a Stock Price Process

- The stock price process is

$$dS = \mu S\, dt + \sigma S\, dz$$

- From Itoˆ 's lemma, it follows that the process followed by a function G of S and t is

$$dG = \left(\frac{\partial G}{\partial S}\mu S + \frac{\partial G}{\partial t} + ½\frac{\partial^2 G}{\partial S^2}\sigma^2 S^2\right)dt + \frac{\partial G}{\partial S}\sigma S\, dz$$

- Both *S* and *G* are affected by the same underlying source of uncertainty, *dz*.

Application to Forward Contracts (p.336)

- Consider a forward contract on a non-dividend-paying stock
- Assume that the risk-free rate of interest is constant and equal to *r* for all maturities.

$$F_0 = S_0 e^{rT}$$

- To know what happens to the forward price as time passes, we define *F* as the forward price at a time *t*, and *S* as the stock price at time *t*, with *t* < T.
- The relationship between *F* and *S* is given by

$$F = Se^{r(T-t)}$$

- Assuming that the stock price, *S* follows the process

$$dS = \mu S\,dt + \sigma S\,dz$$

- We can use Itoˆ ’s lemma to determine the process for *F.* Taking partial derivatives of

$$F = Se^{r(T-t)}$$

- *We get,*

$$\frac{\partial F}{\partial S} = e^{r(T-t)}, \qquad \frac{\partial^2 F}{\partial S^2} = 0, \qquad \frac{\partial F}{\partial t} = -rSe^{r(T-t)}$$

- Substituting these into-

$$dG = \left(\frac{\partial G}{\partial S}\mu S + \frac{\partial G}{\partial t} + \frac{1}{2}\frac{\partial^2 G}{\partial S^2}\sigma^2 S^2\right)dt + \frac{\partial G}{\partial S}\sigma S\, dz$$

- We get the process *F* follows-

$$dF = \left[e^{r(T-t)}\mu S - rSe^{r(T-t)}\right]dt + e^{r(T-t)}\sigma S\, dz$$

Substituting F for $Se^{r(T-t)}$ gives

$$dF = (\mu - r)F\, dt + \sigma F\, dz$$

- Like *S*, the forward price *F* follows geometric Brownian motion with the same volatility as *S* and an expected growth rate of $\mu - r$ rather than μ.

The Lognormal Property

- We now use Itoˆ ’s lemma to derive the process followed by ln S (log of a stock price) when *S* follows the process as before. We define

$$G = \ln S$$

- Since,

$$\frac{\partial G}{\partial S} = \frac{1}{S}, \qquad \frac{\partial^2 G}{\partial S^2} = -\frac{1}{S^2}, \qquad \frac{\partial G}{\partial t} = 0$$

- The process followed by G is

$$dG = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma\, dz$$

- Generalized Wiener process.

- The change in *ln S* between time 0 and some future time *T* is therefore normally distributed as follows,

$$\ln S_T - \ln S_0 \sim \phi\left[\left(\mu - \frac{\sigma^2}{2}\right)T,\ \sigma^2 T\right]$$

or

$$\ln S_T \sim \phi\left[\ln S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T,\ \sigma^2 T\right]$$

- *ln* S_T is normally distributed.
- A variable has a lognormal distribution if the natural logarithm of the variable is normally distributed